Jennifer Brana

jenbrana4@gmail.com | jenniferbrana.github.io | JenniferBrana | jenniferbrana

Research Interests

I am interested in improving the efficiency and programmability of computing systems. I aim to improve how compilers and systems reason about performance to enable efficient execution across diverse architectures.

Research areas: performance modeling & analysis; compilers; computer systems; computer architecture; ML systems.

Education

Carnegie Mellon University — *Pittsburgh, PA*

MS in Computer Science Research — *June 2023 - May 2026*

**Converted PhD admit to masters*

University of Portland — *Portland, OR*

B.S. in Computer Science, *Cum Laude* — *Aug. 2019 - May 2023*

Minor in Computer Engineering.

Refereed Journal Publications

Kobold: Simplified Cache Coherence for Cache-Attached Accelerators — *IEEE CAL 2023*

Jennifer Brana, Brian C. Schwedock, Yatin A. Manerkar, Nathan Beckmann

Refereed Workshop Presentations

Harmony: Co-Optimizing Parallelism and Locality to Bound Performance — *YARCH 2025*

Jennifer Brana, Nathan Beckmann

Kobold: Simplified Cache Coherence for Cache-Attached Accelerators — *Student Research Competition @ MICRO 2022*

Jennifer Brana, Brian C. Schwedock, Yatin A. Manerkar, Nathan Beckmann

Kobold: Simplified Cache Coherence for Cache-Attached Accelerators — *WDDSA @ MICRO 2022*

Jennifer Brana, Brian C. Schwedock, Yatin A. Manerkar, Nathan Beckmann

Honors & Awards

2023 **NSF Graduate Research Fellowship,** National Science Foundation
2023 **Cum Laude,** University of Portland
2023 **Outstanding Student in Computer Science,** University of Portland
2020 **Tau Beta Pi Induction,** Oregon Gamma, University of Portland
All terms **Dean's List,** University of Portland
2019-2023 **President's Scholarship,** University of Portland
2019-2023 **FIRST Robotics Scholarship,** University of Portland

Research Experience

Unaffiliated

Independent Researcher — *August 2026 - Present*

- Characterized performance bottlenecks in SAT solvers. Developing scheduling and runtime techniques to exploit fine- and coarse-grained parallelism in collaboration with Prof. Yatin Manerkar.

Carnegie Mellon University — *Pittsburgh, PA*

Graduate Research Assistant — *June 2023 - May 2026*

- Researching in computer architecture and performance modeling.

AMD Research — *San Jose, CA*

Research Intern — *June 2024 - August 2024*

- Implemented and profiled applications on near-cache accelerators.

Carnegie Mellon University — *Pittsburgh, PA*

Undergraduate Research Assistant — *May 2022 - May 2023*

- Designed and verified cache-coherence protocols for near-cache accelerators. Evaluated performance using architecture simulation.

University of Portland — *Portland, OR*

Undergraduate Researcher — *Jan. 2022 - May 2022*

- Evaluated CPU specialization for Viterbi decoding using architecture simulation, power modeling, and low-level software optimization.

Intelligent, Complex, Adaptive, and Networks Lab — *University of Portland*

Undergraduate Research Assistant — *May 2021 - August 2021*

- Developed software for conducting EEG experiments. Built mechanisms to coordinate and synchronize experimental components.

Service & Leadership

Doctoral Review Committee — *Carnegie Mellon University CSD*

Graduate Student Representative — *Spring 2025 - Spring 2026*

Graduate Student Assembly — *Carnegie Mellon University*

Computer Science Department Representative — *Fall 2023 - Fall 2024*

Tau Beta Pi — *University of Portland*

Oregon Gamma Chapter President — *2021 - 2022*

Mentoring

Research Advising

Amin Hong (CMU ECE master's student) — *Summer 2025*

Bas Yoovidhya (CMU CS master's (thesis) student) — *Fall 2023 - Summer 2024*

Mayne Mei (University of Michigan CS undergraduate, co-advised with Prof. Yatin Manerkar) — *Fall 2023*

Teaching

Carnegie Mellon University

Introductory Course in Leadership and Professionalism (IC-LEAP) — *Instructor, Spring 2025*

University of Portland

Theory of Computation (CS 357) — *Grader, Fall 2022*

Electrical Engineering Dept. Tutor (EE 231, 261, 262, & 332) — *Tutor, 2021-2022*

Logic Design (EE 231) — *Grader, Fall 2021*

Electrical Circuits Lab (EE 271) — *Lab Assistant, Spring 2021*

Skills

Programming C/C++, Python, Java, CUDA, oneTBB, pthreads, RISC-V Assembly, MATLAB, some Haskell and Verilog

Systems & Compilers Linux/Unix, LLVM, gem5, Intel Pin, Linux perf, Murphi Model Checker, Nsight Systems & Compute